

Problem 286: How Time Flies

Difficulty: Medium

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Originally Published: Code Quest 2026

Problem Background

You are part of a Lockheed Martin team managing a critical space mission. The spacecraft's systems log events in different units of time – some in seconds, others in hours or even days. For mission control to analyze the data accurately, you need to standardize the time logs into a specific unit or convert between units as required. Your team has been tasked with building a reliable time conversion tool which will help mission control process and unify time data.

Problem Description

Your task is to write a program that takes an input time value with its units as well as a target unit and calculate the equivalent time in the target unit. The program should handle conversions for seconds, minutes, hours, and days. In this case, all of the inputs and results will be kept to integer values.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include a single line with the following values separated by spaces:

- A positive integer representing the amount of time
- One of "SECONDS", "MINUTES", "HOURS", or "DAYS", representing the time units of the given value
- A different unit from the list above, representing the unit of time to convert into

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3
5 MINUTES SECONDS
60 MINUTES HOURS
7 DAYS HOURS
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Sample Output

For each test case, your program must print a single line in the following format:

- The original amount of time, given as an integer, a space, and the unit in uppercase letters
- A dash and a greater than (>) symbol
- The converted unit of time, given as an integer, a space, and the unit in uppercase letters

You do not need to adjust the units to account for plural values (i.e. “1 HOURS” is a correct output, as shown below).

5 MINUTES->300 SECONDS

60 MINUTES->1 HOURS

7 DAYS->168 HOURS